



Faculty of Engineering

School of Electrical Engineering and Telecommunications

ELEC 1111 – Topic 9

AC Power

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Topic 9 Content

This lecture covers:

- AC power.
 - Instantaneous power.
 - Average power.
 - Maximum average power transfer.
- Effective or RMS value.

**Corresponds to the first part of Chapter 11
of your textbook**

Addressed learning outcomes:

- *LO1*: Systematically analyse DC and AC electric circuits by deriving and solving equations using circuit laws and theorems.
- *LO3*: Apply phasors and sinusoidal steady state analysis to AC circuits.
- *LO6*: Evaluate relevant information to design simple engineering systems that use electrical circuits.

AC power

- **Power analysis** is quite important as all electrical, electronic, and communications systems rely on transmission of power (either in AC or DC form) from one point to another.
- The **most common** form of power is **AC power** due to its low cost and convenience in transmission of high-voltage power from the generator to the consumer.
- There are **multiple concepts** for AC power depending on the application of power and how it is being measured.
- In this lecture just **two main concepts** of AC power are introduced:
 - **Instantaneous** power.
 - **Average** power.

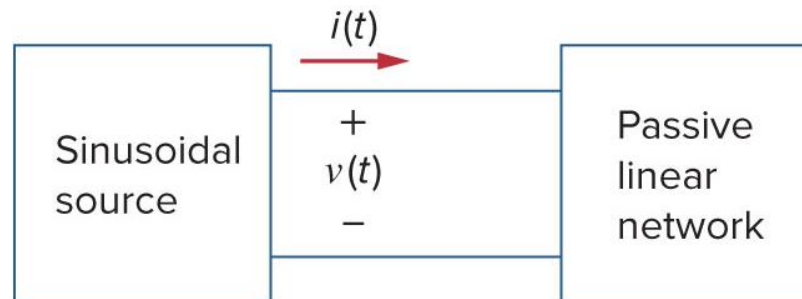


https://en.wikipedia.org/wiki/Transmission_tower

Instantaneous power

- **Instantaneous power** (in watts) is the power at **any instance** of time.
- Instantaneous power $p(t)$ absorbed by an element (or network) is the **product** of the **instantaneous voltage** $v(t)$ across the element (or network) and the **instantaneous current** $i(t)$ through it, assuming **passive sign convention**.

$$p(t) = v(t)i(t)$$



- Instantaneous power is the **rate** at which an element absorbs **energy**.

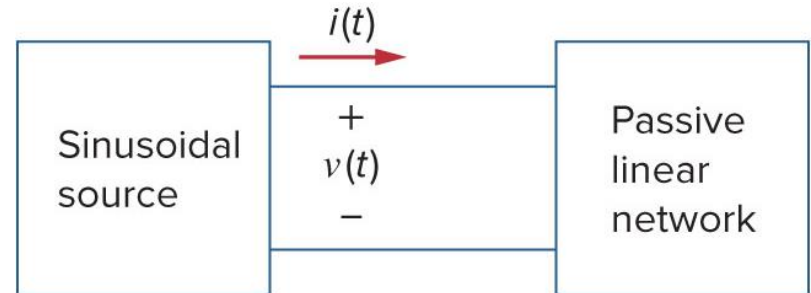
$$w = \int_0^t p(\tau) d\tau \quad \rightarrow \quad p(t) = \frac{dw}{dt}$$

Instantaneous power

- Consider the general case where the voltage and current at the terminals of a circuit with passive elements under sinusoidal excitation are as follows:

$$v(t) = V_m \cos(\omega t + \theta_v)$$

$$i(t) = I_m \cos(\omega t + \theta_i)$$



- The instantaneous power absorbed by the circuit is as follows:

$$p(t) = v(t)i(t) = V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

Apply the trigonometric identity: $\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$

$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

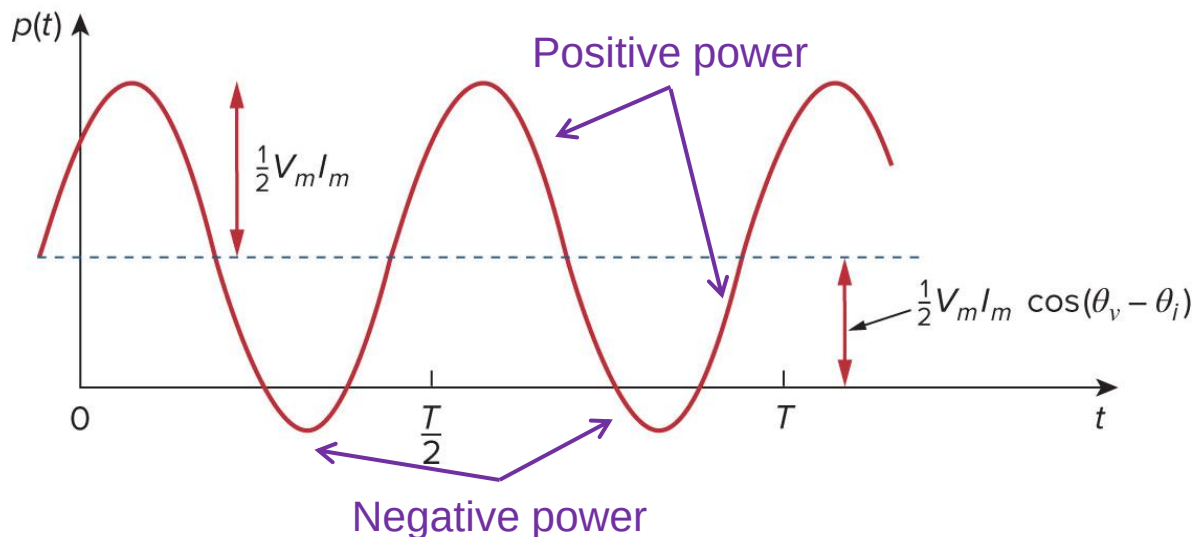
Instantaneous power

- Instantaneous power has two parts:

$$p(t) = \underbrace{\frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)}_{\text{Constant term depending only on phase difference between voltage and current}} + \underbrace{\frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)}_{\text{Sinusoidal term with twice the frequency of voltage or current } T_0 = \frac{T}{2}}$$

Constant term depending **only** on **phase difference** between **voltage** and **current**

Sinusoidal term with **twice** the **frequency** of voltage or current $T_0 = \frac{T}{2}$



- When $p(t)$ is **positive**, power is **absorbed** by the **circuit** (or supplied by the source).
- When $p(t)$ is **negative**, power is **absorbed** by the **source** (or transferred from the circuit to the source).

Note: This is possible due to storage elements (**capacitors** and **inductors**).

Average power

- **Average power** (in watts) is the **average** of the **instantaneous** power over **one period**.
 - Instantaneous power is difficult to measure as it changes with time.
 - The **average power** is more convenient to measure.
 - **Wattmeter**, an instrument for measuring power, provides the **average power**.

$$P = \frac{1}{T} \int_0^T p(t) dt$$

$$P = \frac{1}{T} \int_0^T p(t) dt = \frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) dt + \frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i) dt$$

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) \underbrace{\frac{1}{T} \int_0^T dt}_1$$

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

Integrating a sinusoid over an integer multiple of its period is

zero, $T = 2T_0 = 2 \times \frac{2\pi}{2\omega} = \frac{2\pi}{\omega}$.

The positive area under the sinusoid cancels the negative area

Average power

- **Average power** $P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$ depends on **amplitudes** and **phases** of current and voltage only.
- **Phasor voltage** $\mathbf{V} = V_m \angle \theta_v$ and **phasor current** $\mathbf{I} = I_m \angle \theta_i$ can be used to calculate the average power as follows:

$$\frac{1}{2} \mathbf{V} \mathbf{I}^* = \frac{1}{2} V_m \angle \theta_v \times I_m \angle (-\theta_i) = \frac{1}{2} V_m I_m \angle (\theta_v - \theta_i),$$

where $\mathbf{I}^*$ is the **complex conjugate** of phasor current $\mathbf{I}$.

Transform to rectangular form:

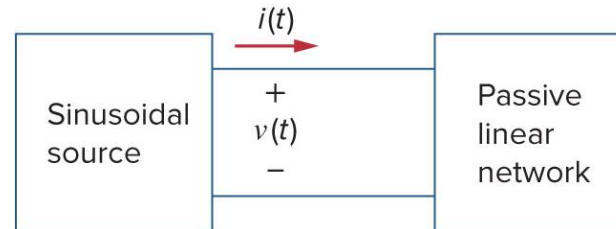
$$\frac{1}{2} \mathbf{V} \mathbf{I}^* = \underbrace{\frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)}_{\text{average power}} + j \frac{1}{2} V_m I_m \sin(\theta_v - \theta_i)$$



$$P = \frac{1}{2} \text{Re}[\mathbf{V} \mathbf{I}^*] = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

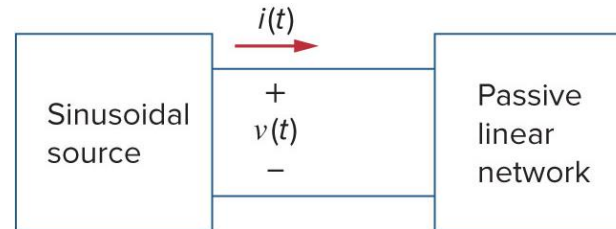
Exercise 1

Given $v(t) = 120 \cos(377t + 45^\circ)$ V and $i(t) = 10 \cos(377t - 10^\circ)$ A, find the instantaneous and average power absorbed by the passive linear network shown below.



Exercise 1

Given $v(t) = 120 \cos(377t + 45^\circ)$ V and $i(t) = 10 \cos(377t - 10^\circ)$ A, find the instantaneous and average power absorbed by the passive linear network shown below.



The instantaneous power is:

$$p(t) = v(t)i(t) = 120 \times 10 \cos(377t + 45^\circ) \cos(377t - 10^\circ)$$

Apply the trigonometric identity: $\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$

$$p(t) = \frac{1}{2} 120 \times 10 \cos(45^\circ - (-10^\circ)) + \frac{1}{2} 120 \times 10 \cos(2 \times 377t + 45^\circ + (-10^\circ))$$

$$p(t) = 600 \cos(55^\circ) + 600 \cos(745t + 35^\circ) \text{ W}$$

The average power is the constant part of $p(t)$: $P = 600 \cos(55^\circ) = 344.2 \text{ W}$

$$\text{or } P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{2} 120 \times 10 \cos(45^\circ - (-10^\circ)) = 600 \cos(55^\circ) = 344.2 \text{ W}$$

Average power

- A **resistive load** $\mathbf{Z} = R$ absorbs **average power** at all times in AC circuits.
- When $\theta_v = \theta_i$, the voltage and current are **in phase**. This implies a **purely resistive** circuit or **resistive load**, $\mathbf{V} = R\mathbf{I}$, thus:

$$P = \frac{1}{2} \operatorname{Re}[\mathbf{V}\mathbf{I}^*] = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

$$P = \frac{1}{2} V_m I_m$$

$$P = \frac{1}{2} R I_m^2 = \frac{1}{2} R |\mathbf{I}|^2$$

$$P = \frac{1}{2} \frac{V_m^2}{R} = \frac{1}{2} \frac{|\mathbf{V}|^2}{R}$$

Proof of the last two:

$$P = \frac{1}{2} \operatorname{Re}[\mathbf{V}\mathbf{I}^*] = \frac{1}{2} \operatorname{Re}[R\mathbf{I} \times \mathbf{I}^*] = \frac{1}{2} R |\mathbf{I}|^2 = \frac{1}{2} R I_m^2$$

$$P = \frac{1}{2} \operatorname{Re}[\mathbf{V}\mathbf{I}^*] = \frac{1}{2} \operatorname{Re}\left[\mathbf{V} \times \frac{\mathbf{V}^*}{R}\right] = \frac{1}{2} \frac{|\mathbf{V}|^2}{R} = \frac{1}{2} \frac{V_m^2}{R}$$

Average power

- A **reactive load** $\mathbf{Z} = j\omega L$ or $\mathbf{Z} = 1/j\omega C$ absorbs **zero average power** in AC circuits.
 - Inductor L and capacitor C absorb zero average power in AC circuits because they **charge and discharge the power** during a full period of the voltage and current.
- When $\theta_v - \theta_i = \pm 90^\circ$, it implies a **purely reactive** circuit or **reactive load**, so:

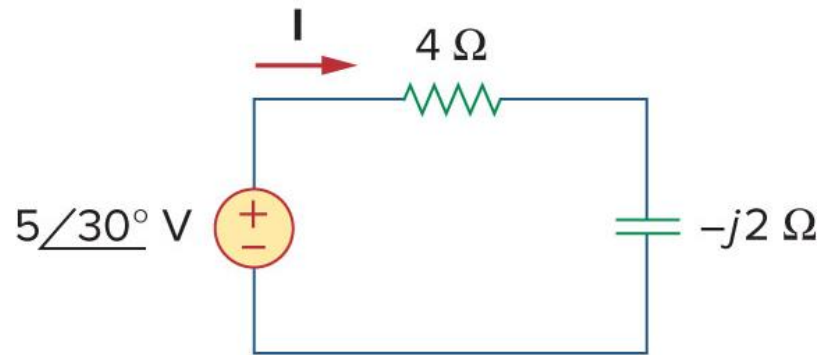
$$P = \frac{1}{2} \operatorname{Re}[\mathbf{VI}^*] = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

$$P = \frac{1}{2} V_m I_m \cos 90^\circ = 0$$

Exercise 2

For the circuit below, find the average power supplied by the source and the average power absorbed by the resistor.

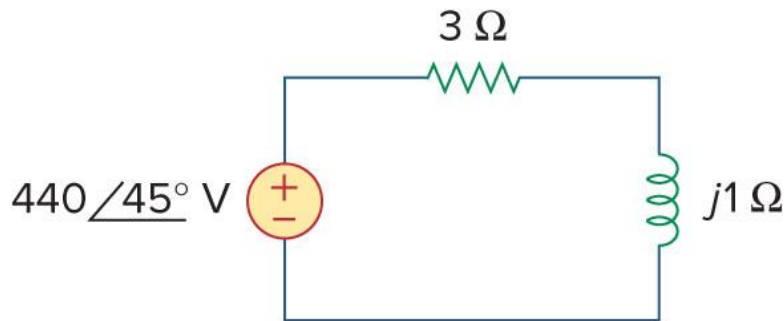
Solution in Exercises_solution_Topic_9.pdf



Exercise 3

For the circuit below, find the average power supplied by the source and the average power absorbed by the resistor and inductor.

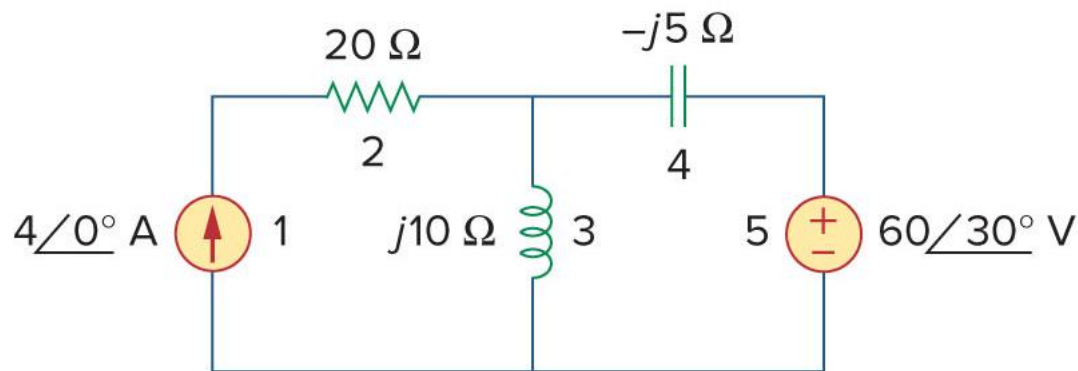
- For practice!
- Answer: 29.04 kW supplied by the source.
29.04 kW absorbed by the resistor.
0 kW absorbed by the inductor.



Exercise 4

Determine the average power generated by each source and the average power absorbed by passive elements in the circuit below.

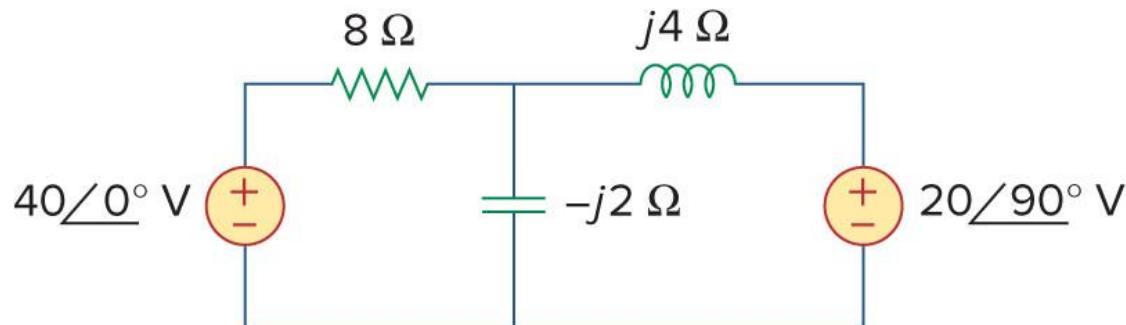
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Exercise 5

Calculate the average power absorbed/supplied by each of the five elements in the circuit below

- For practice!
- Answer: $P_{40V} = 60 \text{ W}$ supplied
 $P_{j20V} = 40 \text{ W}$ supplied
 $P_{8\Omega} = 100 \text{ W}$ absorbed
 $P_{j4\Omega} = P_{-j2\Omega} = 0 \text{ W}$ absorbed



Maximum average power transfer

- Recall that a DC circuit with resistive elements represented by its **Thevenin equivalent circuit** can transfer **maximum power** if and only if $R_L = R_{Th}$.
- The results can be extended to AC circuits in a similar fashion

Consider the Thevenin equivalent circuit shown in Fig. (b) with rectangular form for impedances:

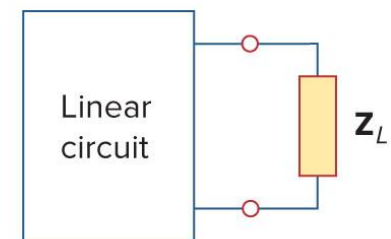
$$\mathbf{Z}_{Th} = R_{Th} + jX_{Th} \quad \text{and} \quad \mathbf{Z}_L = R_L + jX_L$$

The current through the load is:

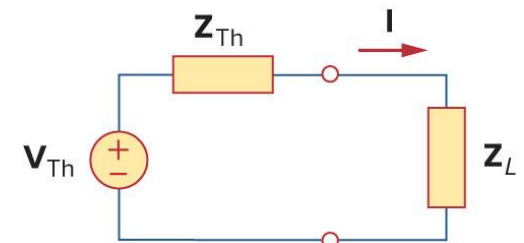
$$\mathbf{I} = \frac{\mathbf{V}_{Th}}{\mathbf{Z}_{Th} + \mathbf{Z}_L} = \frac{\mathbf{V}_{Th}}{(R_{Th} + jX_{Th}) + (R_L + jX_L)}$$

The average power delivered to the load is:

$$P = \frac{1}{2} R_L |\mathbf{I}|^2 = \frac{\frac{R_L |\mathbf{V}_{Th}|^2}{2}}{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2}$$



(a)



(b)

Maximum average power transfer

- To **maximize the average power** in terms of R_L and X_L , set the **partial derivatives** of P with respect to R_L and X_L to **zero** solve them for R_L and X_L :

$$P = \frac{1}{2} R_L |\mathbf{I}|^2 = \frac{\frac{R_L |\mathbf{V}_{Th}|^2}{2}}{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2}$$

$$\frac{\partial P}{\partial X_L} = \frac{-|\mathbf{V}_{Th}|^2 R_L (X_{Th} + X_L)}{[(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2]^2}$$

$$\frac{\partial P}{\partial R_L} = \frac{|\mathbf{V}_{Th}|^2 [(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2 - 2R_L(R_{Th} + R_L)]}{2[(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2]^2}$$

Setting $\frac{\partial P}{\partial X_L}$ and $\frac{\partial P}{\partial R_L}$ to zero results in:

$$X_L = -X_{Th}$$

and

$$R_L = \sqrt{R_{Th}^2 + (X_{Th} + X_L)^2}$$

Combining the results:

$$\mathbf{Z}_L = R_L + jX_L = R_{Th} - jX_{Th} = \mathbf{Z}_{Th}^*$$



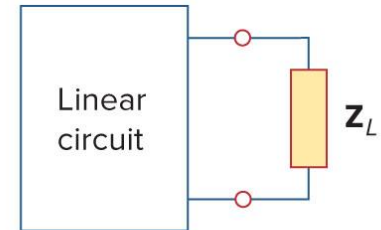
$$\begin{cases} R_L = R_{Th} \\ X_L = -X_{Th} \end{cases}$$

Maximum average power transfer

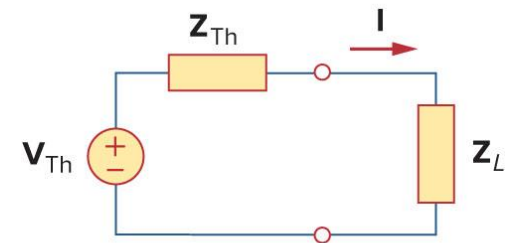
- Setting $R_L = R_{Th}$ and $X_L = -X_{Th}$ in the **average power** equation leads to:

$$\boxed{\mathbf{Z}_L = \mathbf{Z}_{Th}^*} \quad \Rightarrow \quad \boxed{P_{\max} = \frac{|\mathbf{V}_{Th}|^2}{8R_{Th}}}$$

Maximum power is transferred to the load when the **load impedance $\mathbf{Z}_L$** is **equal** to the **complex conjugate** of the **Thevenin impedance $\mathbf{Z}_{Th}^*$** as seen from the load terminals.



(a)



(b)

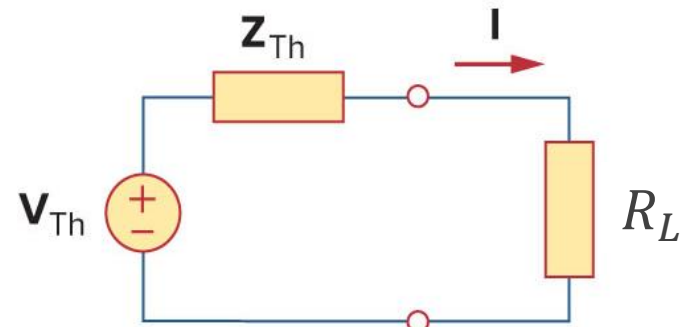
Maximum average power transfer

- For a **pure resistive load** $\mathbf{Z}_L = R_L$, **maximum average power transfer condition** is given by setting $X_L = 0$ in $R_L = \sqrt{R_{Th}^2 + (X_{Th} + X_L)^2}$, as follows:

$$R_L = \sqrt{R_{Th}^2 + X_{Th}^2} = |\mathbf{Z}_{Th}|$$



$$P_{\max} = \frac{1}{2} R_L |\mathbf{I}_L|^2 = \frac{1}{2} \frac{|\mathbf{V}_L|^2}{R_L}$$

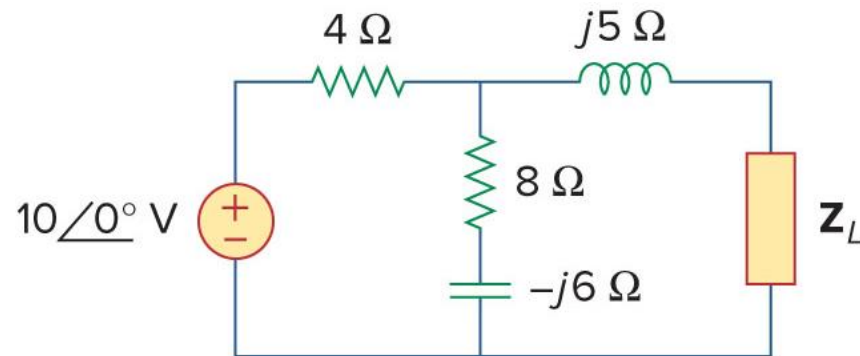


Maximum average power transferred to a **pure resistive load** ($\mathbf{Z}_L = R_L$) can be calculated in an AC circuit using **direct** calculation of **average power** absorbed by a **load resistor**.

Exercise 6

Determine the load impedance $\mathbf{Z}_L$ that maximizes the average power drawn from the circuit below, and then find the maximum average power.

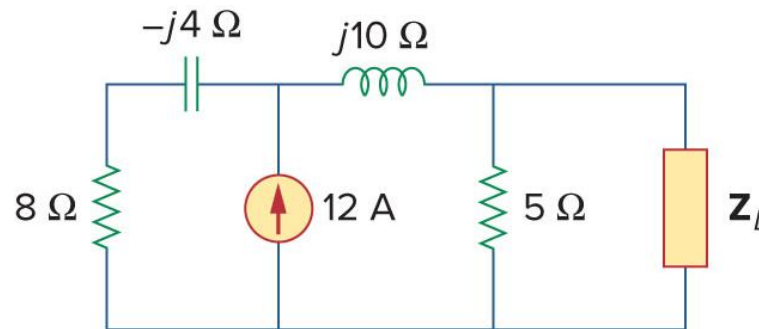
Solution in Exercises_solution_Topic_9.pdf



Exercise 7

For the circuit shown below, find the load impedance $\mathbf{Z}_L$ that absorbs maximum average power. Calculate the maximum average power.

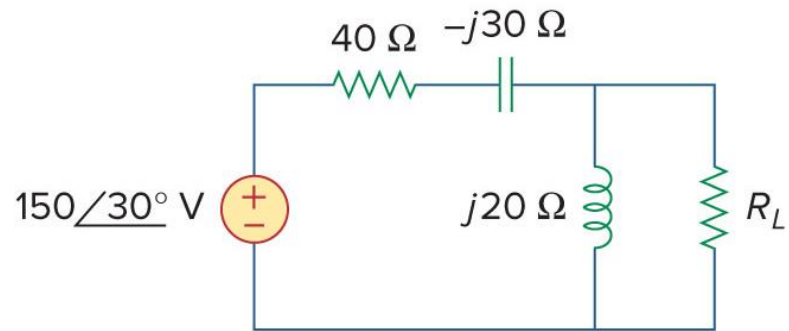
- For practice!
- Answer: $\mathbf{Z}_L = \mathbf{Z}_{Th}^* = 3.415 - j0.731 \Omega$ and $P_{max} = 51.47 \text{ W}$ absorbed



Exercise 8

In the circuit below, find the value of R_L that will absorb the maximum average power, and then calculate that power.

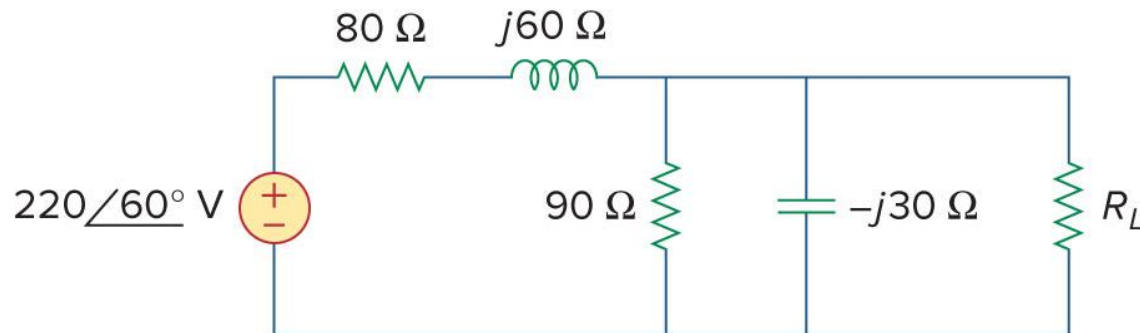
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Exercise 9

For the circuit shown below, find the load resistance R_L such that it absorbs maximum average power, and then calculate the maximum average power transferred to it.

- For practice!
- Answer: $R_L = |\mathbf{Z}_{Th}|^2 = 30 \, \Omega$ and $P_{\max} = 23.06 \, \text{W}$ absorbed



Effective or RMS value

- When a time-varying source is delivering power to a resistive load, we often want to know the effectiveness of that source on delivering power.
- The **effective value** of a **periodic** current is a **DC** current that can deliver **the same average power** to a **resistor** as the periodic current.

Consider the circuit in Fig. (a). The average power is given by:

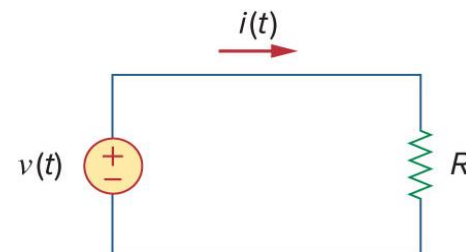
$$P = \frac{1}{T} \int_0^T p(t) dt = \frac{1}{T} \int_0^T Ri(t)^2 dt = \frac{R}{T} \int_0^T i(t)^2 dt$$

The power absorbed in the DC circuit in Fig. (b) is:

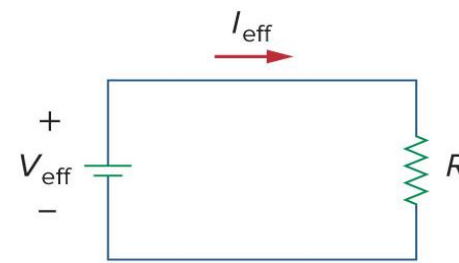
$$P = RI_{\text{eff}}^2$$

The objective is to find I_{eff} such that it delivers the same average power as the AC circuit. Thus:

$$P = RI_{\text{eff}}^2 = \frac{R}{T} \int_0^T i(t)^2 dt \quad \Rightarrow \quad I_{\text{eff}} = \sqrt{\frac{1}{T} \int_0^T i(t)^2 dt}$$



(a)



(b)

Effective or RMS value

- The effective value of the AC voltage is obtained in the same way as the effective current:

$$V_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T v(t)^2 dt}$$

- The effective values for both current and voltage take the form of the **square root** of the **average (mean)** of the **square** of the periodic signal. This is referred to as the **root mean square**, or **RMS value** for short.

$$I_{\text{rms}} = I_{\text{eff}} \quad \text{and} \quad V_{\text{rms}} = V_{\text{eff}}$$

For any periodic signal $x(t)$ in general, the RMS value is given by:

$$X_{\text{eff}} = X_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T x(t)^2 dt}$$

Effective or RMS value

- The RMS value of the **constant signal** (DC signal) is the **constant itself**.
- For a sinusoid $i(t) = I_m \cos(\omega t + \phi)$ the RMS value is obtained as follows:

Using the trigonometric identity
 $\cos^2(a) = \frac{1 + \cos(2a)}{2}$

$$I_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T I_m^2 \cos^2(\omega t + \phi) dt} = \sqrt{\frac{I_m^2}{T} \underbrace{\int_0^T \frac{1}{2} dt}_{= \frac{1}{2}T} + \underbrace{\int_0^T \cos(2\omega t + 2\phi) dt}_{= 0, \text{ Integration over an integer multiple of period}}}$$


 $I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$

- Similarly, for $v(t) = v_m \cos(\omega t + \phi)$  $V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$

Note: These RMS values are only valid for sinusoidal currents and voltages, or sinusoidal signals in general.

Note: An example of RMS value is the **household appliances** working with RMS voltage $V_{\text{rms}} = 220 \text{ V}$, which is:

$$v(t) = 220\sqrt{2} \cos(2\pi 50t) \text{ V}$$

RMS value and average power

- The **average power** can be determined in terms of RMS values:

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{\sqrt{2} \times \sqrt{2}} V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i)$$

$$P = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

- Similarly, the **average power** absorbed by a **resistor** R is given by:

$$P = V_{\text{rms}} I_{\text{rms}}$$

$$P = R I_{\text{rms}}^2$$

$$P = \frac{V_{\text{rms}}^2}{R}$$

Questions?

